

Q1. A differential amplifier is shown in Fig. 1 below, where V_B is a DC voltage. The transistor dimensions are $(W/L)_{1,2}=80$ and $(W/L)_{3,4}=20$. Ignore channel modulation effect for DC calculation and body effect. ($I_0=50\mu A$, $V_{DD}=3V$, $V_{THN}=0.6V$, $V_{THP}=-0.7V$, $\lambda_n=0.1V^{-1}$, $\lambda_p=0.2V^{-1}$, $k'_n=96\mu A/V^2$, $k'_p=32\mu A/V^2$).

- (1) Calculate the differential gain of the amplifier.
- (2) What are the DC voltages at two output nodes when two inputs are biased at a DC voltage with $V_{in}=0$?
- (3) If the output drives a load capacitance of 3pF, calculate the pole frequencies and write the transfer function. (Ignore the parasitic capacitances)

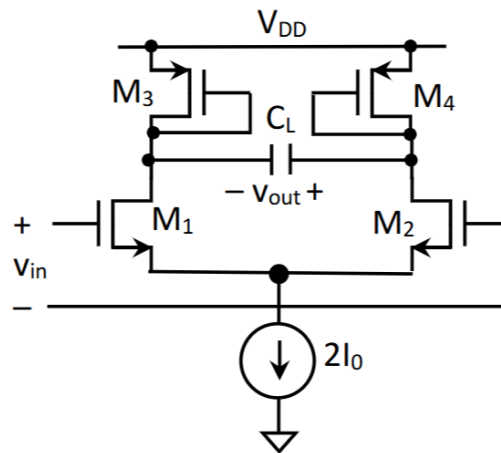


Fig. 1

Solution:

1) Calculate the differential gain of the amplifier

$$A_v = \frac{g_{m1}}{g_{m3}}$$

$$g_{m1} = \sqrt{2k'_n \left(\frac{W}{L}\right)_{M1} I_0} = 876.36 \mu A/V$$

$$g_{m3} = \sqrt{2k'_p \left(\frac{W}{L}\right)_{M3} I_0} = 253 \mu A/V$$

$$\therefore A_v = 3.46$$

Solution:

2) What are the DC voltages at two output nodes when two inputs are biased at a DC voltage with $v_{in} = 0$?

$$I_0 = \frac{k'_p}{2} \left(\frac{W}{L} \right)_{M3} V_{OD}^2 \Rightarrow V_{OD} = 0.395V \Rightarrow V_{SG} = 1.095V$$

$$v_{DC,o1 \text{ or } o2} = 3V - 1.095V = 1.905V$$

Solution:

3) If the output drives a load capacitance of 3pF, calculate the pole frequencies and write the transfer function. (Ignore the parasitic capacitances).

$$w_p = \frac{1}{1/g_{m3} \times 2C_L} = \frac{42.15 \text{Mrad}}{s} = 2 * \pi * 6.7 \text{MHz}$$

$$Av(s) = \frac{Av}{\frac{s}{w_p} + 1} = \frac{3.46}{\frac{s}{2\pi \times 6.7\text{MHz}} + 1}$$

Q2. A differential amplifier is shown in [Fig. 2](#). The gain of amplifier is 60. Determine the sizes of all transistors and DC biases V_{B1} , V_{B2} and input DC bias voltage.

($V_{DD}=3V$, $I_{D5}=0.5mA$, $V_{THN}=0.4V$, $V_{TPN}=-0.64V$, $\lambda_n=\lambda_p=0.04\text{ V}^{-1}$, $k'_n=100\mu A/V^2$, $k'_p=36\mu A/V^2$, and the differential output (V_{out}) swing is $\pm 1V$)

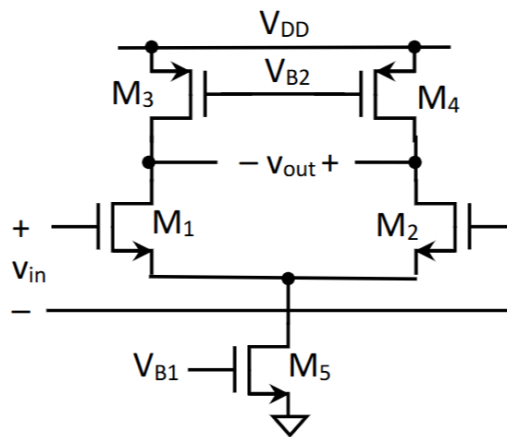
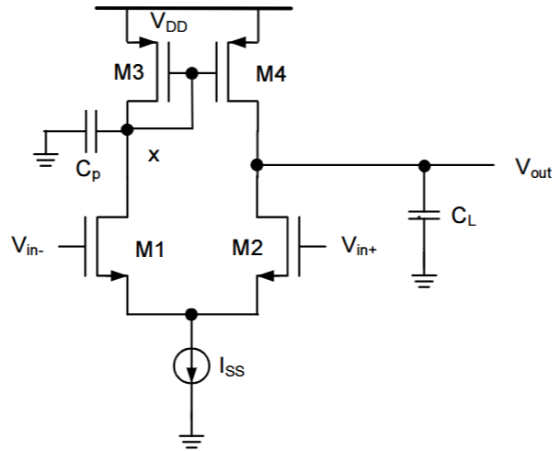


Fig. 2

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Q3. For the differential amplifier shown below, assume the parasitics from transistors can be ignored. ($I_{SS} = 1\text{mA}$, $V_{THN}=0.7\text{V}$, $V_{THP}=-0.8\text{V}$, $\lambda_n=0.1\text{V}^{-1}$, $\lambda_p=0.2\text{V}^{-1}$, $k'_n=134\mu\text{A/V}^2$, $k'_p=38\mu\text{A/V}^2$, $(W/L)_1=(W/L)_2=150$, $(W/L)_3=(W/L)_4=75$, $C_p=10\text{fF}$, and $C_L=0.5\text{pF}$)

- (1) Find the pole and the zero frequencies, and obtain the transfer function.
- (2) Sketch the magnitude and phase response (Bode plot).



Solution:

- 1) Find the pole and the zero frequencies, and obtain the transfer function.

$$g_{m2} = \sqrt{2k'_n \left(\frac{W}{L}\right)_{M2} I_{D2}} = 4.5\text{mA/V}$$

$$g_{m3} = \sqrt{2k'_p \left(\frac{W}{L}\right)_{M3} I_{D3}} = 1.68\text{mA/V}$$

$$r_{o1} = r_{o2} = \frac{1}{\lambda_n I_{D3}} = 20\text{k}$$

$$r_{o3} = r_{o4} = \frac{1}{\lambda_p I_{D3}} = 10\text{k}$$

$$R_{eq1} = r_{o2} || r_{o4} = 6.67\text{k}$$

$$R_{eq2} = r_{o1} || r_{o3} || 1/g_{m3} = 625$$

$$A_{DC} = g_{m2}(r_{o2} || r_{o4}) = 30 \rightarrow 29.5\text{dB}$$

$$\frac{v_{out}(s)}{v_{in}(s)} = \frac{A_{DC}[1 + s/2\omega_{p2}]}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$

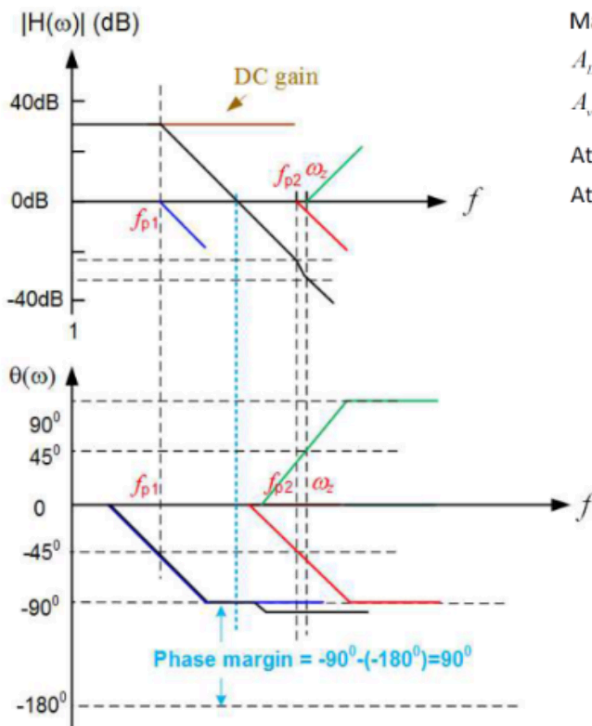
$$A_{DC} = g_{m2}(r_{o2} || r_{o4}) = 30 \rightarrow 29.5dB$$

$$f_{p1} = \frac{\omega_{p1}}{2\pi} = \frac{1}{2\pi R_{eq1} C_L} = \frac{1}{2\pi \times 6.67k \times 0.5pF} = 47.7MHz$$

$$f_{p2} = \frac{\omega_{p2}}{2\pi} = \frac{1}{2\pi R_{eq2} C_p} = \frac{1}{2\pi \times 625 \times 10fF} = 25.5GHz$$

$$f_z = 2f_{p2} = 51GHz$$

- the Req at output is big, much bigger than that at the node X, so the output pole will be dominant
- and the zero is twice that of the second pole, so this amplifier is mainly single pole



Magnitude plot :

$$A_{DC} = 29.5dB \text{ @ } 0 \leq f \leq f_{p1} = 47.7MHz$$

$$A_v = 0dB \text{ @ } f_u = 1.43GHz$$

$$\text{At } f_{p2} = 25.5GHz \rightarrow A_v(f_{p2}) = 0dB - 20\log(25.5GHz / 1.43GHz) \approx -25dB$$

$$\text{At } f_z = 2f_{p2} = 51GHz \rightarrow A_v(f_z) = -25dB - 40\log(51GHz / 25.5GHz) \approx -37dB$$

Phase plot :

Pole1 :

$$\theta(0.1f_{p1} = 4.77MHz) = 0^\circ$$

$$\theta(f_{p1} = 47.7MHz) = -45^\circ$$

$$\theta(10f_{p1} = 477MHz) = -90^\circ$$

Pole2 :

$$\theta(0.1f_{p2} = 2.55GHz) = -90^\circ$$

$$\theta(f_{p2} = 25.5GHz) = -135^\circ$$

$$\theta(10f_{p2} = 255GHz) = -180^\circ$$

Zero @ LHP $\rightarrow$ leading phase) :

$$\text{For } 0.1f_{p2} \leq f \leq 0.1f_z = 5.1GHz,$$

$$\theta = -90^\circ - 45^\circ \log(5.1GHz / 2.55GHz) = -103.5^\circ$$

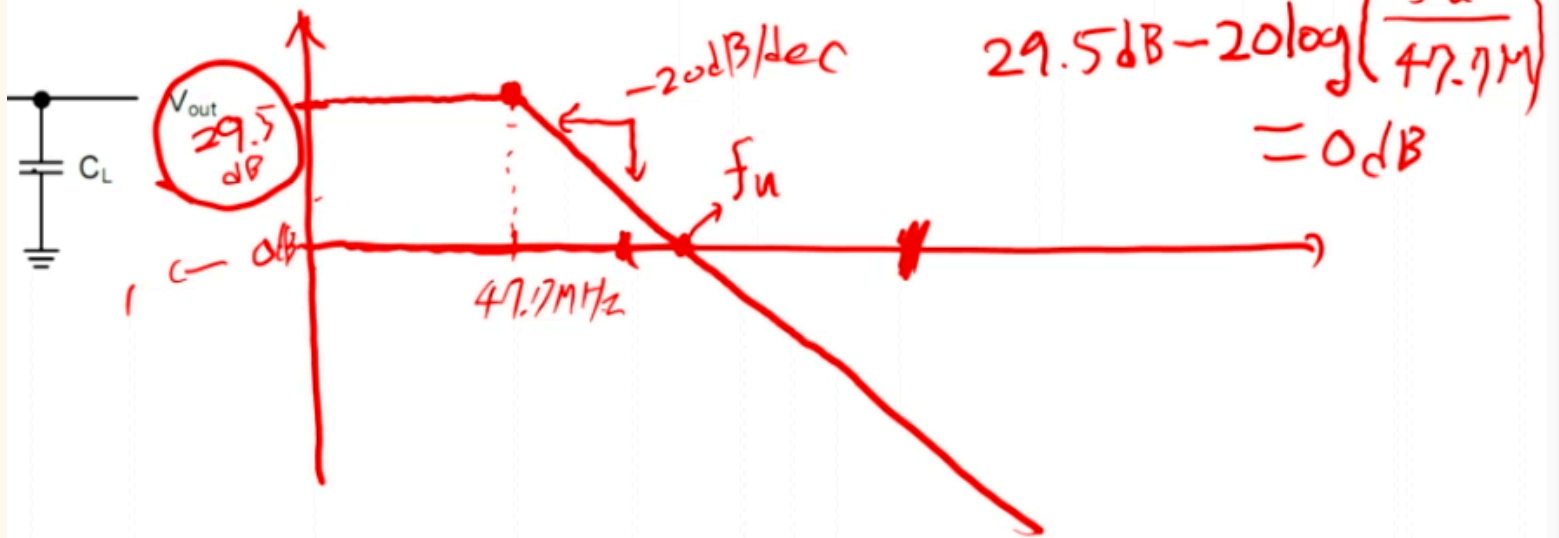
$$\text{For } f > 0.1f_z \text{ (} f_{p2} \text{ and } f_z \text{ cancelled),}$$

$$\theta = -103.5^\circ$$

Question 3

Solution:

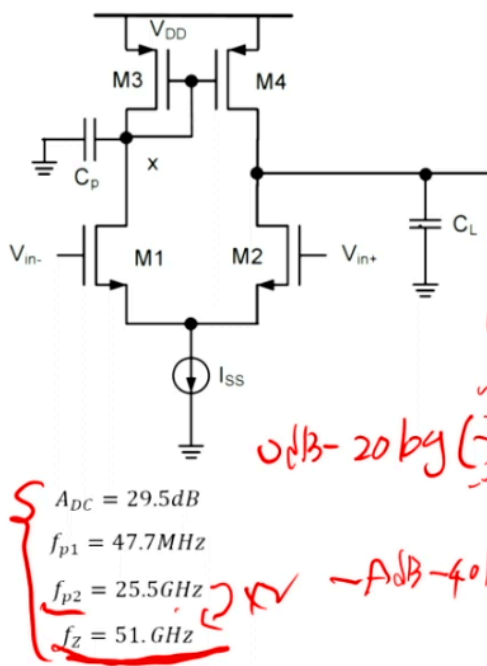
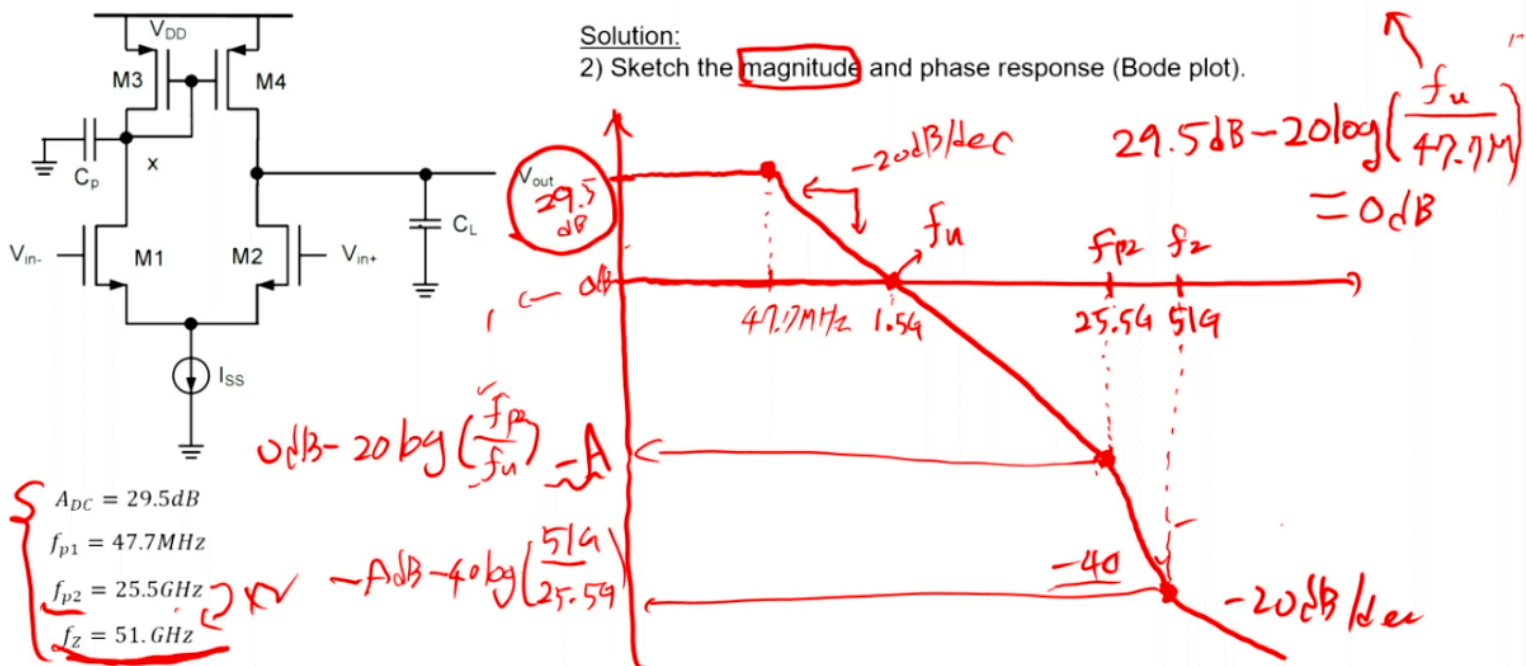
2) Sketch the magnitude and phase response (Bode plot).



Question 3

Solution:

2) Sketch the magnitude and phase response (Bode plot).



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- can estimate where the second pole is before or after the 0dB point by using -20dB/decade to estimate
- the other magnitude can be estimated also
 - put in log, because if its 10 times, it give -20

Magnitude plot :

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$$\text{At } f_{p2} = 25.5GHz \rightarrow A_v(f_{p2}) = 0dB - 20\log(25.5GHz / 1.43GHz) \approx -25dB$$

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Phase plot :

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$$\text{For } f > 0.1f_z \text{ (} f_{p2} \text{ and } f_z \text{ cancelled),}$$

$$\theta = -103.5^\circ$$

Pole2 :

$$\theta(0.1f_{p2} = 2.55GHz) = -90^\circ$$

$$\theta(f_{p2} = 25.5GHz) = -135^\circ$$

$$\theta(10f_{p1} = 255GHz) = -180^\circ$$

- here is the calculations